Assignment 5

Submitted By: Nischay Maharjan

Continue the EDA on the Google Playstore Data to find the answers to the following questions:

1. Which Category has the largest number of installations??

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```python
#Calculating the top categories by grouping the Category column, summing the total installs for each category, and selecting the top 7.

top_categories = (
    df_copy.groupby("Category")["Installs"].sum()
            .sort_values(ascending=False)
            .head(7)
            .reset_index()
)

top_categories.head(3)
```

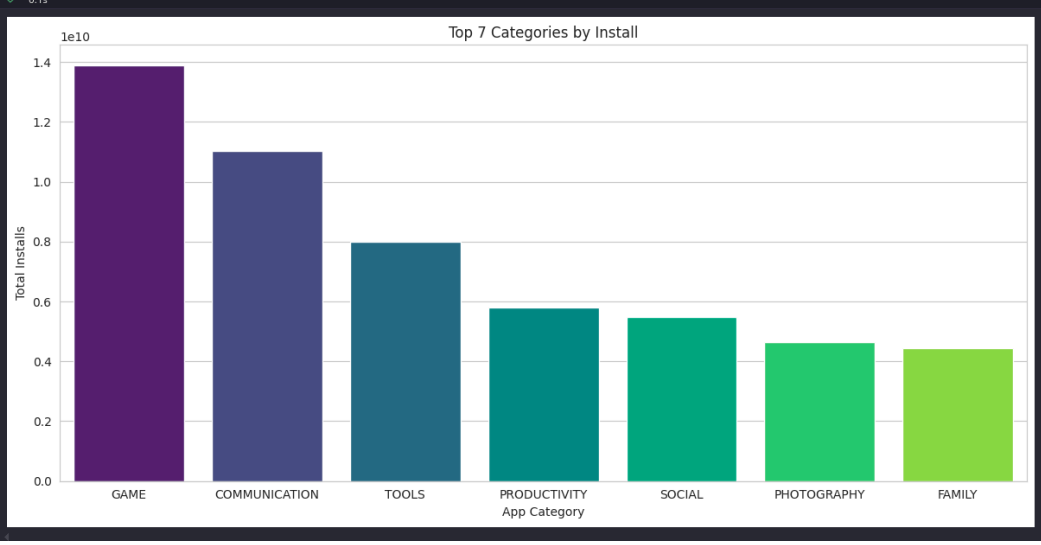
[297] ✓ 0.0s

	Category	Installs
0	GAME	13878924415
1	COMMUNICATION	11038276251
2	TOOLS	8001771915

```python
#Plotting barchart to see the top 7 category with largest number of installation

plt.figure(figsize=(12, 6))
sns.barplot(data=top_categories, x="Category", y="Installs", palette="viridis")
plt.xlabel("App Category")
plt.ylabel("Total Installs")
plt.title("Top 7 Categories by Install")
plt.tight_layout()
```

[298] ✓ 0.1s



2. What are the Top 5 most installed Apps in Each popular Categories ??

```python
#Getting top 5 categories
top_categories = (
    df_copy.groupby("Category")["Installs"].sum().sort_values(ascending=False).head(5).index
)

top_category_df = df_copy[df_copy["Category"].isin(top_categories)]
```

[300] ✓ 0.0s

```python
#Using for loop for each category and filtering out the top 5 most installed apps and appending to list which is then later converted to dataframe
top_apps_list=[]

for category in top_categories:
    category_df = df_copy[df_copy["Category"] == category]
    top_5_apps = category_df.sort_values("Installs", ascending=False).head(5)

    display(f"Category: {category}") #better table formatting
    display(top_5_apps)

    top_apps_list.append(top_5_apps)

top_apps_df = pd.concat(top_apps_list).reset_index(drop=True)
```

[301] ✓ 0.1s

'Category: GAME'

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver	LU_Day	LU_Month
1654	Subway Surfers	GAME	4.5	27722264	76000.0	1000000000	Free	0.0	Everyone 10+	Arcade	2018-07-12	1.90.0	4.1 and up	12	7
1662	Pou	GAME	4.3	10485308	24000.0	500000000	Free	0.0	Everyone	Casual	2018-05-25	1.4.77	4.0 and up	25	5
1722	My Talking Tom	GAME	4.5	14891223	32000.0	500000000	Free	0.0	Everyone	Casual	2018-07-19	4.8.0.132	4.1 and up	19	7
1661	Temple Run 2	GAME	4.3	8118609	62000.0	500000000	Free	0.0	Everyone	Action	2018-07-05	1.49.1	4.0 and up	5	7
1655	Candy Crush Saga	GAME	4.4	22426677	74000.0	500000000	Free	0.0	Everyone	Casual	2018-07-05	1.129.0.2	4.1 and up	5	7

'Category: COMMUNICATION'

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Curr
340	Gmail	COMMUNICATION	4.3	4604324	32000.0	1000000000	Free	0.0	Everyone	Communication	2018-08-02	Varies wit
335	Messenger – Text and Video Chat for Free	COMMUNICATION	4.0	56642847	32000.0	1000000000	Free	0.0	Everyone	Communication	2018-08-01	Varies wit
338	Google Chrome: Fast & Secure	COMMUNICATION	4.3	9642995	32000.0	1000000000	Free	0.0	Everyone	Communication	2018-08-01	Varies wit
336	WhatsApp Messenger	COMMUNICATION	4.4	69119316	32000.0	1000000000	Free	0.0	Everyone	Communication	2018-08-03	Varies wit
341	Hangouts	COMMUNICATION	4.0	3419249	32000.0	1000000000	Free	0.0	Everyone	Communication	2018-07-21	Varies wit

'Category: TOOLS'

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver
3234	Google	TOOLS	4.4	8033493	32000.0	1000000000	Free	0.0	Everyone	Tools	2018-08-03	Varies with device
3265	Gboard - the Google Keyboard	TOOLS	4.2	1859115	32000.0	500000000	Free	0.0	Everyone	Tools	2018-07-31	Varies with device
7536	Security Master - Antivirus, VPN, AppLock, Boo...	TOOLS	4.7	24900999	32000.0	500000000	Free	0.0	Everyone	Tools	2018-08-04	4.6.6
3255	SHAREit - Transfer & Share	TOOLS	4.6	7790693	17000.0	500000000	Free	0.0	Everyone	Tools	2018-07-30	4.5.28_ww
4005	Clean Master- Space Cleaner & Antivirus	TOOLS	4.7	42916526	32000.0	500000000	Free	0.0	Everyone	Tools	2018-08-03	Varies with device

'Category: PRODUCTIVITY'

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver	L
3454	Google Drive	PRODUCTIVITY	4.4	2731171	32000.0	1000000000	Free	0.0	Everyone	Productivity	2018-08-06	Varies with device	Varies with device	
3476	Google Calendar	PRODUCTIVITY	4.2	858208	32000.0	500000000	Free	0.0	Everyone	Productivity	2018-08-06	Varies with device	Varies with device	
3473	Dropbox	PRODUCTIVITY	4.4	1861310	61000.0	500000000	Free	0.0	Everyone	Productivity	2018-08-01	Varies with device	Varies with device	
3574	Cloud Print	PRODUCTIVITY	4.1	282460	32000.0	500000000	Free	0.0	Everyone	Productivity	2018-05-23	Varies with device	Varies with device	
3450	Microsoft Word	PRODUCTIVITY	4.5	2084126	32000.0	500000000	Free	0.0	Everyone	Productivity	2018-07-11	16.0.10325.20043	4.4 and up	

'Category: SOCIAL'

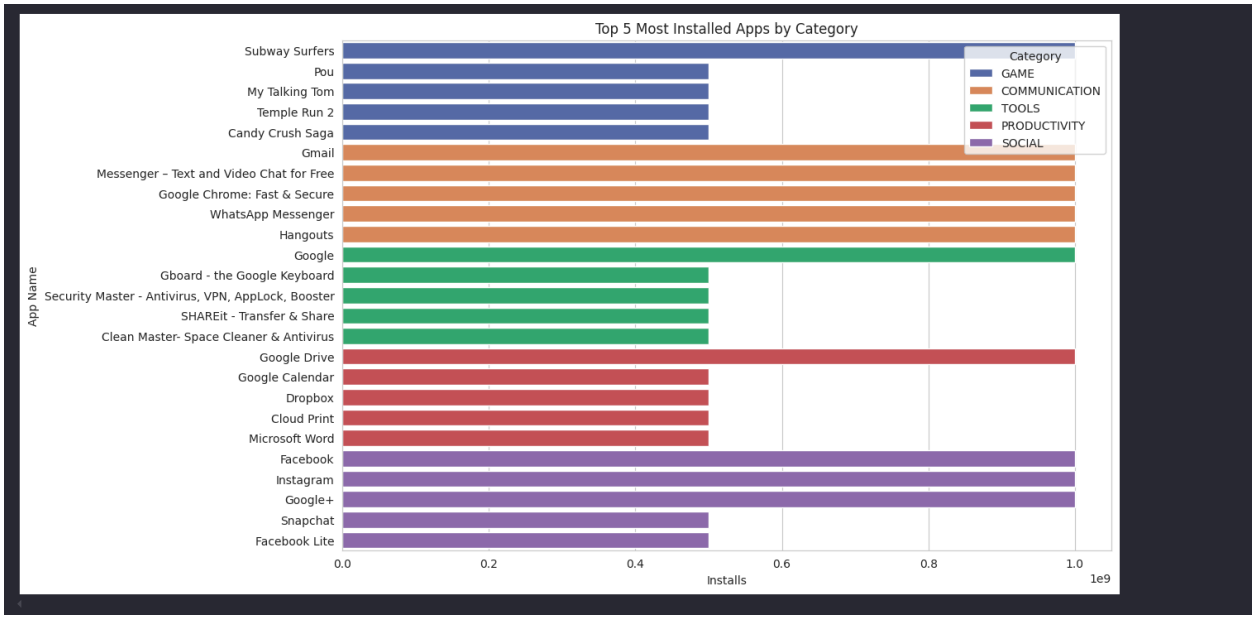
App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver	LU_Day	LU

```python
plt.figure(figsize=(12, 8))

sns.barplot(data=top_apps_df, x="Installs", y="App", hue="Category", palette="deep"
)

plt.title("Top 5 Most Installed Apps by Category")
plt.xlabel("Installs")
plt.ylabel("App Name")
```

[302] ✓ 0.5s



3. How many apps are there on Google Play Store which get 5 ratings??

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```
rating_df = df_copy[df_copy["Rating"] == 5]
rating_df
```

[303] ✓ 0.0s

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver	LU_Day	LU_Month	LU_Year
329	Hojiboy Tojiboyev Life Hacks	COMICS	5.0	15	37000.0	1000	Free	0.0	Everyone	Comics	2018-06-26	2.0	4.0.3 and up	26	6	2018
612	American Girls Mobile Numbers	DATING	5.0	5	44000.0	1000	Free	0.0	Mature 17+	Dating	2018-07-17	3.0	4.0.3 and up	17	7	2018
615	Awake Dating	DATING	5.0	2	70000.0	100	Free	0.0	Mature 17+	Dating	2018-07-24	2.2.9	4.4 and up	24	7	2018
633	Spine- The dating app	DATING	5.0	5	93000.0	500	Free	0.0	Teen	Dating	2018-07-14	4.0	4.0.3 and up	14	7	2018
636	Girls Live Talk - Free Text and Video Chat	DATING	5.0	6	50000.0	100	Free	0.0	Mature 17+	Dating	2018-08-01	8.2	4.0.3 and up	1	8	2018
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
10721	Mad Dash Fo' Cash	GAME	5.0	14	16000.0	100	Free	0.0	Everyone	Arcade	2017-06-19	2.5a	4.1 and up	19	6	2017
10742	GKPB FP Online Church	LIFESTYLE	5.0	32	79000.0	1000	Free	0.0	Everyone	Lifestyle	2017-12-31	0.7.1	4.4 and up	31	12	2017
10776	Monster Ride Pro	GAME	5.0	1	24000.0	10	Free	0.0	Everyone	Racing	2018-03-05	2.0	2.3 and up	5	3	2018
10820	Fr. Daoud Lamei	FAMILY	5.0	22	86000.0	1000	Free	0.0	Teen	Education	2018-06-27	3.8.0	4.1 and up	27	6	2018
10837	Fr. Mike Schmitz Audio Teachings	FAMILY	5.0	4	36000.0	100	Free	0.0	Everyone	Education	2018-07-06	1.0	4.1 and up	6	7	2018

271 rows × 16 columns

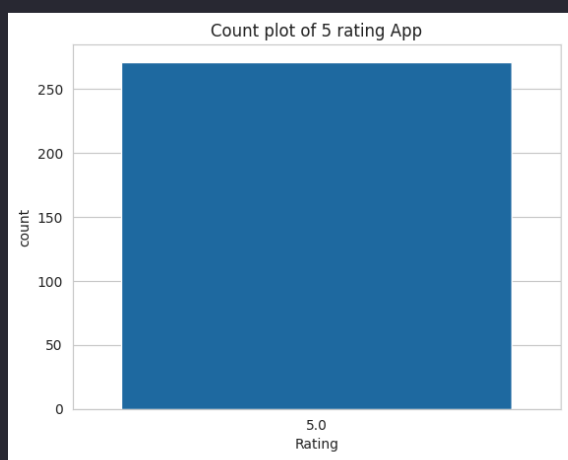
```
num_apps_with_5_rating = len(rating_df)
print(f"There are {num_apps_with_5_rating} apps with 5 rating")
```

[304] ✓ 0.0s

```
There are 271 apps with 5 rating
```

```
sns.countplot(data=rating_df,x=rating_df["Rating"])

plt.title("Count plot of 5 rating App")
```

[305] ✓ 0.1s

```
Text(0.5, 1.0, 'Count plot of 5 rating App')
```